PROGRAMMING

Java Interview Questions & Concepts

Core Java, OOP & Memory Architecture

1. Object-Oriented Principles in Java

- Encapsulation: Restricting direct access to object fields using private access and public getters/setters.
- Inheritance: Code reusability via 'extends'; Java supports single class inheritance, multiple interface inheritance.
- Polymorphism: Compile-time (Method Overloading) vs. Runtime (Method Overriding with dynamic dispatch).
- Abstraction: Abstract classes (can have state/constructors) vs. Interfaces (pure contract, default methods since Java 8).

2. JVM Internals & Memory Model

- JVM Components: ClassLoader Subsystem, Runtime Data Areas, Execution Engine (JIT Compiler & Garbage Collector).
- Memory Areas: Heap (shared, objects live here, Eden/Survivor/Tenured) vs. Stack (per-thread, frames, local primitives).
- Garbage Collection: Mark-and-Sweep, G1GC, ZGC. 'System.gc()' merely suggests collection to the JVM.
- String Pool: String immutability enables caching in PermGen/Metaspace String Constant Pool.

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Collections, Concurrency & Modern Java

3. Collections Framework Deep Dive

- List: ArrayList (dynamic array, fast random access) vs. LinkedList (node pointers, fast insertion at ends).
- Set: HashSet (backed by HashMap, O(1) average lookup) vs. TreeSet (Navigable, sorted by Red-Black Tree, O(log n)).
- Map: HashMap internal working: Array of buckets, hash collision handled via LinkedList/Treeify threshold of 8.
- ConcurrentHashMap: Thread-safe without synchronizing entire table (Segment/bucket level locking).

4. Multi-Threading & Java 8+ Features

- Creating Threads: Extending Thread class vs. Implementing Runnable/Callable interface.
- Thread States: NEW, RUNNABLE, BLOCKED, WAITING, TIMED_WAITING, TERMINATED.
- Functional Interfaces: SAM (Single Abstract Method) annotated with @FunctionalInterface (Predicate, Function, Consumer, Supplier).
- Stream API: Declarative pipelines: filter(), map(), flatMap(), reduce(), collect(). Lazy evaluation via terminal operations.